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Digital Data in Qualitative Research: Strengthening the Visibility and Credibility of Portraiture

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Abstract

This study investigates how various methods of using digital technologies can support qualitative study. These technologies impact the research processes of data collection, data conversion, data organization, data analysis, and presentation of results (Bruce & Levin, 1998; Fetterman, 2002). Examples of three qualitative studies are shared that incorporated the gathering, analysis, and integration of digital data. Findings indicate that incorporating digital resources can provide “e-readers” with evidence relating to issues of methodologies, researcher intrusiveness, participant voice, and context. These benefits combine to strengthen the genre of portraiture, by making the processes and products of this type of qualitative study more public and open to evaluation (Anfara, Brown, & Mangione, 2002).

Introduction

While qualitative studies are heralded for their collections of rich data and their abilities to enhance our understandings of the dynamics embedded in complex educational settings, they are often suspect, as well. Consumers of the research often question the trustworthiness of the data presented, relying heavily on the researcher’s expertise in defining themes and identifying the most salient examples on which to draw conclusions. Researchers have been accused of possessing *blind spots* in matters of using media to convey their evidence and assertions (Fischman, 2001). While the genre of portraiture has been defended convincingly as a credible form of educational research (Lawrence-Lightfoot & Davis, 2002), weaknesses must be addressed in this type of practice (English, 2000). Challenges remain concerning the intrusiveness, or non-intrusiveness, of the researcher, accuracy of their descriptions, and the appropriateness or contextual relevance of the data selected for inclusion in final presentations (Peshkin, 2000). In the genre of portraiture, where the researcher endeavors to paint a picture of an educational setting or phenomenon, these challenges require a solid investment of rigor and integrity.

Analyzing data from multiple sources (i.e. a palette of interviews, observations, documents, photographs, etc.) and blending these elements into a product that captures the essence of the environment under investigation are difficult tasks. Traditional qualitative methods including triangulation and member checking play a crucial role in focusing the image, but researchers still struggle to provide readers with enough evidence to convince them of the trustworthiness of the representation.

As qualitative methodologies have been evolving, the world of information technologies has been expanding, as well. Communication systems including the World Wide Web, electronic databases, and digital multimedia repositories are making it possible for educational researchers to share their work in multiple venues and formats. The development of hypertext, portable document files (pdf), digital media

(image, audio, and video), and various interactive and easily accessible digital publishing options (www, CD-ROM, DVD) provide researchers with new and exciting ways to practice their craft. The following paragraphs describe how developers of portraiture might use digital resources to strengthen and deepen their work, utilizing these resources. The descriptions also include digital artifacts from final research products in web-based, CD-ROM and DVD formats.

Hypertext and PDFs

In the past it was not feasible to include full transcriptions of interview or observation notes as appendices in qualitative studies (and some would argue that this type of disclosure is still not desirable). Such inclusion could conceivably add hundreds of pages of text to a research report. Researchers had to be trusted to select the evidence that most convincingly supported their arguments and consumers of the research were left to judge if the task was well done. However, documents transcribed digitally can now be made available in small, easily transferable files. Word-processed documents can be saved in a variety of formats, including hypertext markup language (html) and portable document format (pdf) for digital distribution. A research report can include hyperlinks that connect the reader to the full transcription, giving the consumer access to the complete interview or observation document, for their review. Such access provides an avenue for researchers to more openly share their discoveries and for consumers of the research to participate in the process of data analysis, if they choose.

Digital Media

Though reading an interviewee's exact words can convey a great deal of meaning, the text alone lacks a complete representation of the participant's response. Capturing and transferring the audio or video taped version of the interview (or observation) and making these elements available to the reader provides a more distinct presentation of the experience. Again, this type of data-sharing was once difficult, if not impossible. While audio and video formats have been available for decades, only recently have these elements been able to be captured in digital form, making them as easily linkable and transferable as hypertext. Digital technologies now allow this information to be converted and linked to the final research report in various ways.

Incorporating audio and video data into the research design helps to extend the potential of thick description. When images, and audio and video clips were shared with the teachers participating in the cited studies, or colleagues interested in the research, they were very positive in [their responses](#). This data can also be used as verification for the descriptions presented in the text. Actually hearing the emotion and inflection in a teacher or student's voice as they share their ideas, or seeing the students working in the computer lab helps to support the written descriptions in the research reports.

Besides providing alternate ways of reviewing data, audio and video excerpts can give consumers of research opportunities to see and hear the researcher in action. Such opportunities allow for greater scrutiny of the impact the researcher, herself, had in the research setting. As a scholar listens to an excerpt from an interview, they can analyze the relationship of the interviewer and interviewee, based on the tone and inflection present in the voices. Hearing the pauses, sighs, and chuckles embedded in an interview session can bring a greater sense of authenticity to the descriptions and conclusions of the researcher. In a similar way, reviewing video footage that includes the researcher in the environment under study allows scholars to make judgments about the intrusiveness of the researcher. How comfortable do the participants seem in the presence of the researcher? Does it appear some might be presenting a façade rather than going about their business as usual? These are questions that can be addressed when video and audio data are included in a final research product.

Digital Publishing

Granting the consumers of qualitative research multiple ways of reviewing the data and the research methodologies has the potential of strengthening the trustworthiness of the research, as well as making

the research more visible, interesting and interactive. Several formats can be harnessed by qualitative researchers to enhance their work. Beginning in a hypertext environment, scholars can begin to construct portraits that are more compelling descriptions. A digital document that contains links to observation and interview transcriptions, as well as bibliographic information, allows a reader to explore the study in more depth. Such an environment also gives the reader more choice in how the research is examined and evaluated. While this type of presentation is still text-based, the hypertext format allows for a more substantive inclusion of the data and options for interactivity.

Once a researcher begins to utilize the properties of a hypertext environment, she can then pursue the addition of media elements to her portrait products. Inserting images and linking digital audio and video files can be the next step in painting a more intricate and multi-dimensional description of the environment under study. Once these elements are introduced, the writer can also look to webpage design and CD-ROM/DVD development to build research narratives that allow readers to more completely examine the connections of the data and researcher extrapolations. In this way, exploring the possibilities offered through digital technologies can give qualitative researchers a broader and deeper palette from which to create their portraits.

Examples of Digital Data in Portraiture

The following paragraphs synthesize the work of one qualitative researcher over a four-year period; the analysis is the result of a self-study process (Bullough & Pinnegar, 2001). Journal notes, work logs, and transcriptions for research presentations provide the data in this investigation, as a teacher-researcher model was used (Burdell & Swadener, 1999; Hubbard & Power, 1999). Three qualitative studies incorporating digital technologies in collecting, converting, categorizing, analyzing and presenting data are examined. The studies' methodologies and practices are scrutinized, focusing on how these inquiries were expanded and enhanced with digital data. Frustrations and concerns about implementing these practices are discussed.

Three final research products are used to illustrate and support the assertions of this study, and excerpts from these products are included within this manuscript. A web-based, refereed journal article (Banister, 2002), dissertation CD-ROM (Banister, 2001) and a research study on DVD (Banister, 2003) provide pieces of illustration and evidence for the conclusions drawn. Incorporation of digital, hyperlinked data throughout a research manuscript addresses English's (2000) concerns about independent access to information from a qualitative study. Providing full transcriptions of interview and observation notes, images of the research setting, scanned documents related to the study, and audio and video vignettes open up a portrait-type study for alternative explanations. These elements also can present a more public view of the researcher's role, addressing questions of positions of power and voice.

Portraiture in HyperText

This study examined a particular school improvement initiative within a small, Midwestern elementary school. The complete electronic journal version of this study is located at http://www.education.uiowa.edu/jrel/spring02/Banister_0108.htm. This article includes only minimal examples of hyperlinking, providing the reader with links to bibliographic and footnote information, only. This functionality is problematic to some degree because the user, once directed to the information, cannot easily return to the body of the manuscript. Inserting a *Back* link that would return to the point of departure would make this feature more user-friendly. However, the potential to use this structure to link to external web resources and supporting media elements is apparent. One of the most useful characteristics of this type of publication is the access to a broad readership. Additions of email or discussion forum features to this environment would provide an infrastructure for continued dialogue about the descriptions and findings presented.

This example represents the first step in the evolutionary process of moving from print media into hypertext environments. The publication, while web-based, is still functioning in a fairly static state.

However, even the process of this translation extended the impact of the writing. Despite the lack of email and discussion forum features in this publication, the work still prompted three e-readers to seek out the author via email with questions and comments regarding the manuscript. While similar responses might have occurred from the article in traditional print media format, the hypertext setting facilitates this type of dialogue by providing tools that are easy to use and accessible.

Expanding the dialoguing capabilities within an online journal can provide a venue for a richer exchange in exploring a qualitative study. The archiving feature of a discussion forum allows for continuity in a discussion that might evolve over months or even years; such a process is not easily replicated in static print situations. The analysis of the history of the dialogue could be commentary that enriches the scholarship and understanding of the phenomenon under study. Once the characteristics of hypertext are employed to enhance qualitative studies such as portraiture, exploring other types of digital possibilities follows swiftly.

Portraiture in HyperText with CD-ROM

Extending the hypertext environment to include additional media elements is modeled in the author's dissertation study. While the multiple case study was published as a traditional, hardcopy volume (Banister, 2001), a CD-ROM appendix was included that gave readers additional data and resources related to the work. The CD included the complete hypertext version of the dissertation, with links to various documents and media elements cited (see Images 1, 2, and 3). Full interview (127 pages) and observation (66 pages) transcriptions (in pdf format) were also provided. In addition, an interactive interface allowed readers to view the text, by chapters, with accompanying video and audio elements. The disk contained over 400 pages of text, 36 images, and over an hour of audio and video footage, arranged in an interactive format that allowed users the ability to explore the data and narratives in a variety of ways.

Media and textual elements were combined to give readers a more descriptive experience of the case studies. For example, this type of narrative/observation/interview combination is demonstrated in the third grade case study when the topic of technical support for classroom computers was explored (Video 1).



Video 1: Technical Support for Classroom Computers

Juxtaposing video footage of the kindergarten students during center-time and the audio transcription of the kindergarten teacher's interview about computer use during center-time also provides the reader with a uniquely-triangulated description (Video 2).



Video 2: Computer Use during Center Time

In addition, the reader also sees and hears the researcher during observation and interview settings, providing the opportunity to evaluate the researcher's impact on the environment. Audio elements, integrated into the research narrative provide readers with a better sense of context, as well. The [fatigued inflection of a third graders voice](#) as she talks about the time required to create a word-processed document or a [teacher's passionate response](#) concerning technology's relationship to other aspects of schooling are examples.

Expanding the hypertext environment to include full-text transcriptions allows readers to visit the complete context of an interview or observation experience. This ability provides an element of openness that was impossible to achieve before digital text files could be incorporated. The addition of these transcriptions gives the reader access to raw data, and judgments about coding, themes and interpretations can be made.

Sharing these data sources more openly with colleagues provides resources for those teaching and learning about qualitative research methodologies. The CD-ROM described here was used by a professor teaching qualitative methods (Figure 1). She reported that her students were challenged by exploring and analyzing the data contained on the disk. Because of the digital form of the data (pdf files), students could be given almost 400 pages of text as a small, email attachment. The text, then, could be analyzed, coded and compared with classmates. The lively classroom discussions that ensued from this activity reflected the authentic nature of the experience.

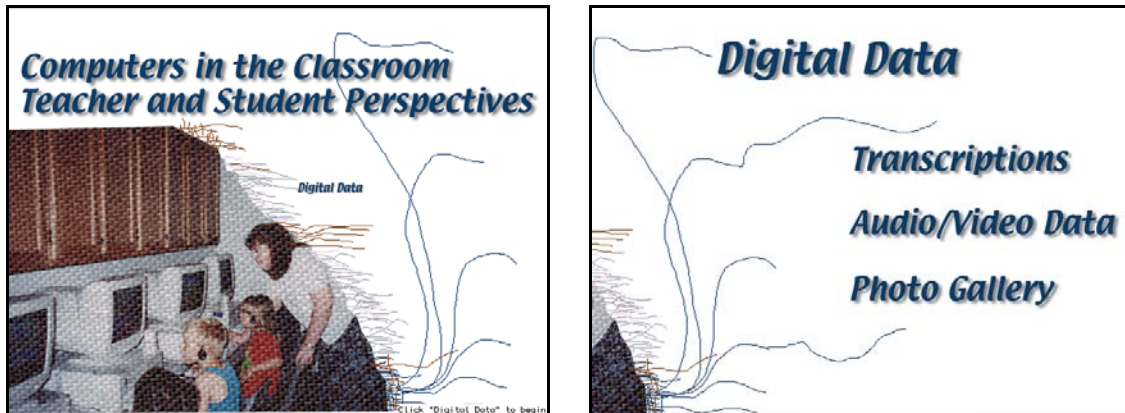


Figure 1: Qualitative Research Methods CD-ROM Screenshots

In the genre of portraiture, visual and auditory data truly enhance the presentation of the phenomenon portrayed. While textual descriptions, if well done, can give readers an image that is compelling, the addition of digital graphics, video and audio create a portrait that is deeper and richer than the text alone (Figure 2). Combining these elements adds a dimension of realism that strengthens the trustworthiness of the text. The practice of weaving together the audio from an interview, with video footage of an event being discussed, while simultaneously including the researcher's written interpretations, creates a vibrant research artifact. *E-readers* respond to these representations with interest and insights into the complexities of the research enterprise.



Figure 2: Use of Various Media to Portray a Phenomenon

Portraiture in DVD

More advanced digital technologies can be tapped to create a “high-tech” genre of interactive video portraiture. This type of research report can be illustrated from a narrative embedded in DVD format (Banister, 2003). This particular example consists of a 45-minute video that paints a picture of a fifth grade classroom’s integration of computer technology (Figure 3). Teacher and student interviews, as well as observation footage, is combined with the researcher’s narration to tell a story of an exemplary teacher (Video 3). A separate hypertext document links to various elements of the DVD and within the DVD structure, users can navigate to additional documents (student activity sheets, rubrics, etc.). Such an environment allows *e-readers* to analyze the data presented and the researcher’s position within the research setting to better critique the quality of the final assertions.

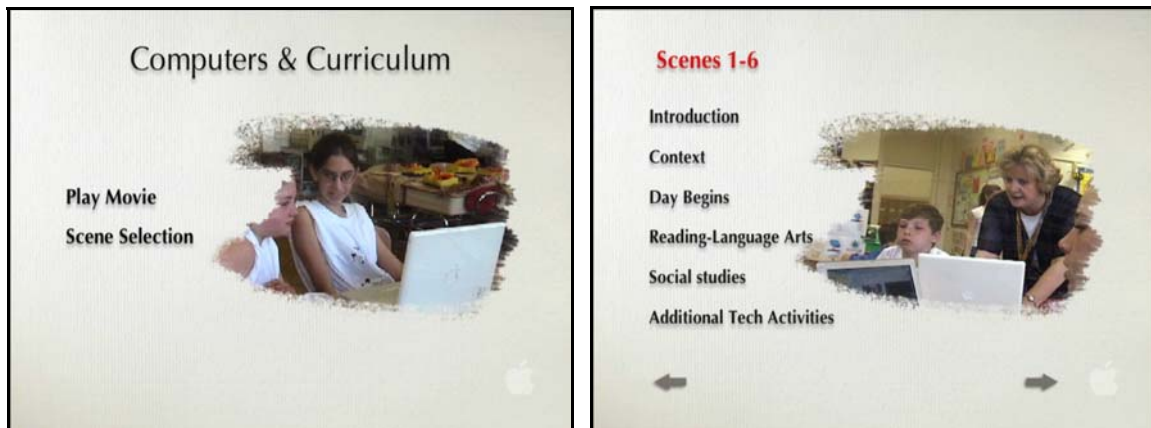
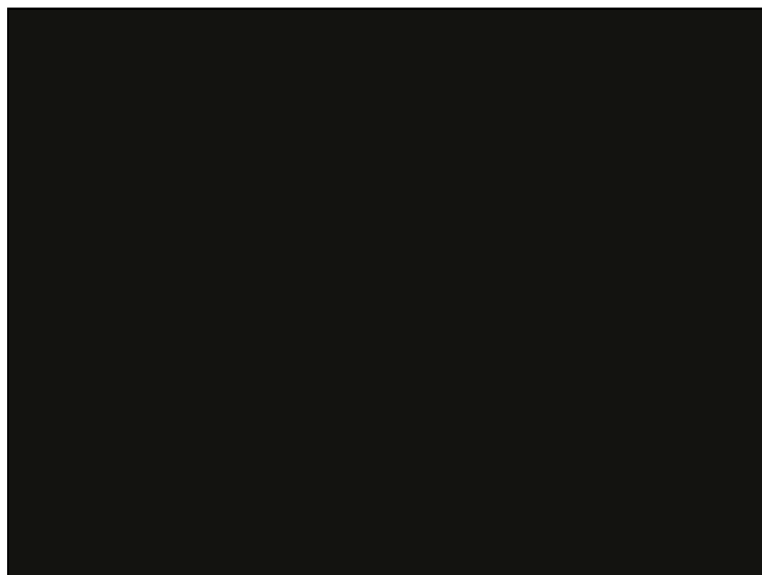


Figure 3: Narrative in DVD Format



Video 3: Video That Tells a Story

The venue of DVD has the potential of taking portraiture to a new level. Rather than read the researcher’s descriptive synthesis of a phenomenon, *e-readers* can view the phenomenon, as it unfolds. Video-taped interviews and researcher’s commentaries can be woven into the final product, and supporting

documents can be linked, in addition. This format provides the *e-reader* with a rich, three-dimensional experience. However, assembling a quality DVD research portraiture is not an easy task.

Coding and editing video footage is time-consuming and the researcher must remain committed to sharing the data that relates most to the research question(s), not the video data that is most visually appealing or anecdotally entertaining. Of course, the audio and video quality of the media is important, but the story to be told should be the priority. In the particular DVD cited here, excerpts exist where classroom noises nearly overpower the interviewee's response or where the classroom lighting is not optimal for video taping. The researcher struggles with capturing authentic representations of a phenomenon clearly, without allowing the technology of cameras, microphones, and staged lighting to disrupt and intrude upon the environment under study.

Implications

Digital data, connected seamlessly into qualitative research reports, have potential for strengthening the genre of portraiture. Giving the audience access to the context (e.g., complete transcriptions) allows for further examination and interpretation of the data. Video and audio elements from the study provide evidence of the researcher's relationship/influence in the setting (i.e., non-intrusiveness, intimidation, etc). With this type of data the audience is also able to critique the researcher's accuracy in descriptions of the setting. Hyperlinking all of this information allows for non-linear exploration and analysis of ideas presented in the narrative.

Data Collection

While embracing the use of digital data in qualitative work is promising, researchers need to consider several factors when implementing such an approach. From the inception of the project, planning must be completed with an eye toward the final product. Issues in collection of audio, images and video for digital environments must be addressed; good quality in capturing these elements initially will insure quality in the final products. Participant permissions and issues of intrusion, equity, and democracy must be considered (Eisenhart, 2001). Several months before data collection begins, the researcher needs to specifically identify what types of data will be sought (i.e., audio, video, VR, image). Once these elements are chosen, negotiation with the participants on the parameters of such data collection and written permission articulating these agreements should be developed. Next, the equipment needed to collect the data, in the various formats, must be secured. The researcher may need additional technical support in locating and operating the equipment required. Finally, plans to save and catalog the digital material collected must be confirmed and implemented.

Data Conversion

Qualitative researchers need to develop a comfort level with hypertext and digital image, audio and video technologies to successfully integrate digital data into their work. Issues of operating systems, file types, file management, hardware and software must be considered. Some may want to find a colleague or assistant with these skills to deal with the practical concerns of this process.

Data Organization and Analysis

Due to issues of file size, all audio and video footage cannot be included in the final research product. Decisions about themes and relevance to theory development still need to be addressed. However, because *e-readers* are able to compare media elements with full-text observation and interview transcriptions, greater assurances of accuracy of context and interpretation are available. Researchers must expand their enlightened eyes (Eisner, 1998) to critique and edit multiple formats of data. Organizing these elements may be similar to the code and retrieve framework of digitally managing qualitative text data (Crowley, Harre, & Tagg, 2002; Richards & Richards, 1998). The development of

connoisseurship in this new genre is important and edits also require a sensitivity to ethical responsibilities, such as intrusion and voice (Rosenstein & Sheva, 2002).

Data Presentation

Careful consideration must be given to the final format(s) a research report will take. Will it be an electronic narrative which includes hyper links to elements of digital data? Will other forms of appendices be included that are digital in nature? Will the culminating portraiture be published to the web, CD-ROM, DVD, or other venue? Will cumulative records, such as full transcriptions, be included? Creativity and intuitiveness in design of the user interface is key. A substantial amount of time, equipment, expertise, and technical support will be required to complete a strong study (Rogers, 2000). The normal messiness of qualitative work will be increased during this process, and the focus must be kept on the story, not on the technology.

Conclusions

Up to this point, most uses of computer technologies in qualitative study have concentrated on data management and retrieval. This study demonstrates that there are powerful ways for qualitative researchers to integrate hyperlinked environments to produce final research products that strengthen and enhance the field. Incorporating digital data into qualitative designs of portraiture can produce a richer, and more complex, representation. The extended and additional information passed on to the reader allows for greater trustworthiness. Providing the audience with these resources creates a setting for increased engagement, translating to greater visibility of research questions and assertions. Greater visibility generates several benefits for educational researchers. Reaching wider audiences with qualitative research findings increases the likelihood that those findings will have an impact on the improvement of educational systems and classrooms. These possibilities should encourage us to explore and pursue the integration of digital data in our inquiries.

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